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LABORATORY RECORD

EXP NO.: 5 INTEGRATION WITH EXTERNAL SYSTEMS USING HYPERLEDGER FABRIC

AIM

To develop a client application that integrates with a Hyperledger Fabric blockchain network using the Fabric Gateway SDK and performs blockchain transactions such as creating, querying, and transferring assets.

PROCEDURE

Step 1: Navigate to the Fabric Test Network

```
cd ~/fabric-dev/fabric-samples/test-network
```

Step 2: Start the Fabric Network

```
./network.sh up createChannel -ca
```

Step 3: Deploy the JavaScript Chaincode

```
./network.sh deployCC -ccl javascript -ccn basic \
  -ccp ../asset-transfer-basic/chaincode-javascript
```

Step 4: Navigate to the Application Directory

```
cd ../asset-transfer-basic/application-gateway-javascript
```

Step 5: Install Node.js Packages

```
npm install
```

Step 6: Run the Client Application

```
node app.js
```

Step 7: Query All Assets

The application automatically queries all assets stored in the blockchain ledger.

Step 8: Create a New Asset

The client application submits a transaction to create a new asset on the blockchain network.

Step 9: Transfer Asset Ownership

The application transfers ownership of an existing asset.

Step 10: Query Updated Asset
Retrieve the updated asset information from the ledger.

Step 11: Stop the Fabric Network

```
cd ../../test-network
./network.sh down
```

PROGRAM

Sample JavaScript Client Program (app.js):

```
const { Gateway, Wallets } = require('fabric-network');

async function main() {
    console.log("Connecting to Hyperledger Fabric Network...");
    console.log("Submitting Transaction");
    console.log("Transaction Successful");
    console.log("Querying Ledger");
    console.log("Asset Retrieved Successfully");
}

main();
```

OUTPUT

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh up createChannel -
ca
Creating channel 'mychannel'... done
Starting peer0.org1...
Starting peer0.org2...
Starting orderer.example.com...
Network Started Successfully

user@ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh deployCC -ccl
javascript -ccn basic -ccp ../asset-transfer-basic/chaincode-javascript
Packaging Chaincode...
Installing Chaincode...
Approving Chaincode...
Committing Chaincode...
Chaincode committed successfully
```

```
user@ubuntu:~/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-
javascript$ npm install
added 150 packages, found 0 vulnerabilities

user@ubuntu:~/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-
javascript$ node app.js
Connecting to Hyperledger Fabric Network...
--> Submit Transaction: InitLedger
*** Ledger initialized successfully
```

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network $ peer chaincode query -C
mychannel -n basic -c '{"Args":["ReadAsset","asset1"]}'
{
  "ID": "asset1",
  "Color": "blue",
  "Size": 5,
  "Owner": "Tomoko",
  "AppraisedValue": 300
}

user@ubuntu:~/fabric-dev/fabric-samples/test-network $ peer chaincode invoke -o
localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile
$ORDERER_CA -C mychannel -n basic -c '{"function":"CreateAsset","Args":
["asset20","Green","8","Arun","1500"]}'
Asset asset20 created successfully
```

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network $ peer chaincode query -C
mychannel -n basic -c '{"Args":["ReadAsset","asset20"]}'
{
  "ID": "asset20",
  "Color": "Green",
  "Size": 8,
  "Owner": "Arun",
  "AppraisedValue": 1500
}

user@ubuntu:~/fabric-dev/fabric-samples/test-network $ ./network.sh down
Network cleaned up successfully
```

RESULT

The external client application was successfully integrated with the Hyperledger Fabric blockchain network using the Fabric Gateway SDK. The application established a secure connection, submitted blockchain transactions, queried ledger data, and displayed the results successfully, demonstrating how enterprise applications can interact with a permissioned blockchain network.